

# Eye-tracking visual attention during MRI safety instructions

Hans Marien, Marijn Schraagen, Henk Aarts

short paper ACM 14th International Conference on Human-Agent Interaction (HAI 2026)

NLP Group Meeting September 10th, 2026



# Literacy levels in the Netherlands

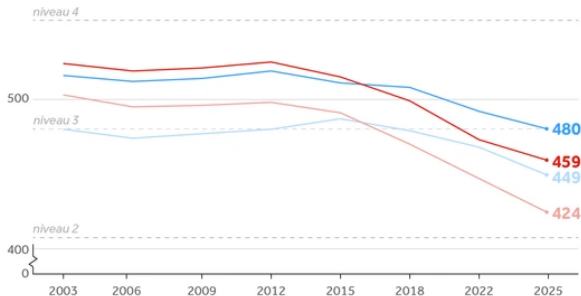
- Literacy levels are in decline
- Level 2 “minimum level to participate in society”

## Leesvaardigheid Nederlandse scholieren daalt

Gemiddelde leesvaardigheidsscore

● NL meisjes ● NL jongens ● EU-14 meisjes ● EU-14 jongens

600 punten



Bron: PISA

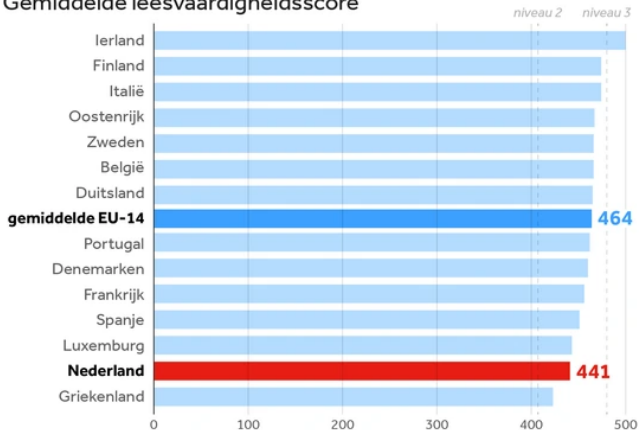


# Literacy levels in the Netherlands

- Low scores compared to other European countries

## Nederland scoort lager dan de meeste EU14-landen

Gemiddelde leesvaardigheidsscore



Bron: PISA



# Health literacy

- General low literacy contributes to low health literacy
- “The degree to which individuals have the capacity to obtain, process, and understand basic healthcare information and services needed to make appropriate health decisions”
- Misunderstanding healthcare instructions or advice can have serious consequences



# Modality of healthcare instructions

- Use case: patients preparing to enter an MRI scanner
- Risk of injury or failed scan when wearing metals or moving inside the scanner
- Modality of information: written, spoken, video, talking to human healthcare provider
- When human is not available: use of **avatar**
- Different modalities lead to different levels of understanding



# Avatar pilot



Pilot with low-literate patients and control group: text, speech, face movement

# Eye-tracking

- Reading difficulties are visible in gaze behavior
  - Longer fixations
  - Re-reading
  - Visual attention allocation
- Attention shifts between written text and avatar face
- Measuring with eye-tracking



# Research goal

- Examine visual attention to verbal/nonverbal expressions of the avatar face and accompanying written text
- Measure differences between low literate participants and control group





# Methodology

- Participants: 23 limited reading skills, 10 with typical reading skills
- Task: watch MRI safety instructions and answer comprehension questions
- Six segments with instructions
  - Synchronized spoken and written information
  - Nonverbal face movements
- Record eye movement in two Areas of Interest (Aoi): avatar face and text

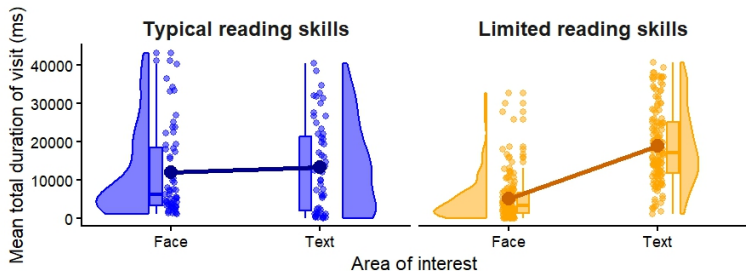


# Methodology



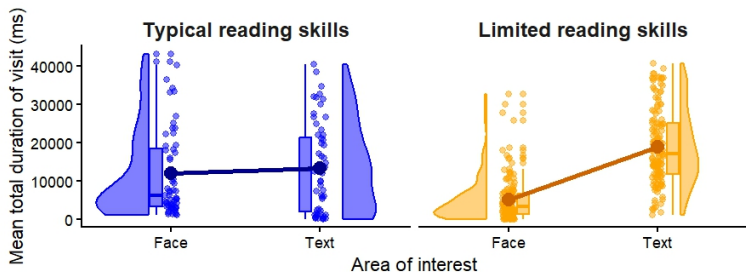
Texts designed to be understandable, confirmed with readability formula for Dutch

# Results



- Typical reading skills: equal attention between face and text
- Limited reading skills: more attention to text
- Differences in score on comprehension questions: 5.5/6 vs. 6/6
- Statistical significance for all observations

# Interpretation



- Limited reading skills: more attention to text
- More time needed to understand the content
- Post-task comprehension lower: simple text with high readability scores is not sufficient for understanding
- Avatar did not attract much attention in both groups

# Implications

- Literacy levels can/should be considered when using text and avatars
- Potentially, low-literacy can be detected using eye tracking
  - Less intrusive than language proficiency test
- Interface that adapts to user attention and comprehension possible
  - Simple, small and relatively cheap eye-tracking device attached to monitor



# Adaptation using eye-tracking

## The Pupil Becomes the Master: Eye-Tracking Feedback for Tuning LLMs

Samuel Kiegeland<sup>1</sup> David R. Reich<sup>2</sup> Ryan Cotterell<sup>1</sup> Lena A. Jäger<sup>2,3</sup> Ethan Wilcox<sup>1</sup>

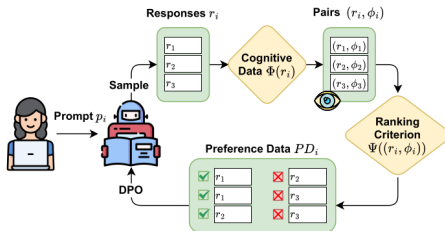


Figure 1. Framework for using cognitive data such as eye-tracking data for tuning LLMs. We sample responses from a model and obtain cognitive data for them. Each pair of responses is ranked using various criteria, such as minimizing fixation durations resulting in a preference dataset for DPO. See §3 for details.



# Limitations

- Alternative explanations for the results
  - differences in trust of avatars between groups
  - systematic differences in unrelated variables such as age
- Human healthcare professionals giving information may be preferred
- Various directions for future research

